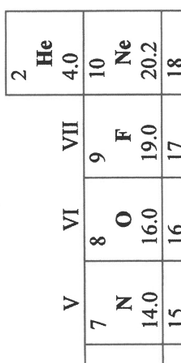
PERIODIC TABLE 周期表



GROUP 族

I	II	III	IV	V	VI	VII	0
							2 **He** 4.0
3 **Li** 6.9	4 **Be** 9.0	5 **B** 10.8	6 **C** 12.0	7 **N** 14.0	8 **O** 16.0	9 **F** 19.0	10 **Ne** 20.2
11 **Na** 23.0	12 **Mg** 24.3	13 **Al** 27.0	14 **Si** 28.1	15 **P** 31.0	16 **S** 32.1	17 **Cl** 35.5	18 **Ar** 40.0
19 **K** 39.1	20 **Ca** 40.1	21 **Sc** 45.0	22 **Ti** 47.9	23 **V** 50.9	24 **Cr** 52.0	25 **Mn** 54.9	26 **Fe** 55.8
37 **Rb** 85.5	38 **Sr** 87.6	39 **Y** 88.9	40 **Zr** 91.2	41 **Nb** 92.9	42 **Mo** 95.9	43 **Tc** (98)	44 **Ru** 101.1
55 **Cs** 132.9	56 **Ba** 137.3	57 * **La** 138.9	72 **Hf** 178.5	73 **Ta** 180.9	74 **W** 183.9	75 **Re** 186.2	76 **Os** 190.2
87 **Fr** (223)	88 **Ra** (226)	89 ** **Ac** (227)	104 **Rf** (261)	105 **Db** (262)			

*	58 **Ce** 140.1	59 **Pr** 140.9	60 **Nd** 144.2	61 **Pm** (145)	62 **Sm** 150.4	63 **Eu** 152.0	64 **Gd** 157.3
**	90 **Th** 232.0	91 **Pa** (231)	92 **U** 238.0	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)

27 **Co** 58.9	28 **Ni** 58.7	29 **Cu** 63.5	30 **Zn** 65.4			
45 **Rh** 102.9	46 **Pd** 106.4	47 **Ag** 107.9	48 **Cd** 112.4	49 **In** 114.8	50 **Sn** 118.7	51 **Sb** 121.8	52 **Te** 127.6	53 **I** 126.9	54 **Xe** 131.3
77 **Ir** 192.2	78 **Pt** 195.1	79 **Au** 197.0	80 **Hg** 200.6	81 **Tl** 204.4	82 **Pb** 207.2	83 **Bi** 209.0	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)

65 **Tb** 158.9	66 **Dy** 162.5	67 **Ho** 164.9	68 **Er** 167.3	69 **Tm** 168.9	70 **Yb** 173.0	71 **Lu** 175.0
97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (260)

Marking Schemes

This document was prepared for markers' reference. It should not be regarded as a set of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret its content with care.

Chemistry
Paper 1

SECTION A

Question No. Part I	Key		Question No. Part II	Key	
1.	A	(70%)	25.	B	(52%)
2.	C	(91%)	26.	A	(76%)
3.	B	(80%)	27.	B	(72%)
4.	D	(81%)	28.	D	(66%)
5.	C	(56%)	29.	C	(53%)
6.	B	(63%)	30.	B	(64%)
7.	A	(73%)	31.	A	(77%)
8.	A	(66%)	32.	C	(49%)
9.	B	(74%)	33.	A	(46%)
10.	C	(77%)	34.	C	(69%)
11.	B	(49%)	35.	D	(27%)
12.	D	(67%)	36.	D	(60%)
13.	C	(65%)			
14.	B	(76%)			
15.	A	(80%)			
16.	A	(62%)			
17.	C	(52%)			
18.	D	(82%)			
19.	D	(36%)			
20.	C	(76%)			
21.	D	(37%)			
22.	D	(71%)			
23.	B	(61%)			
24.	A	(81%)			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

General Marking Instructions

1. In order to maintain a uniform standard in marking, markers should adhere to the marking scheme agreed at the markers' meeting.

2. The marking scheme may not exhaust all possible answers for each question. Markers should exercise their professional discretion and judgment in accepting alternative answers that are not in the marking scheme but are correct and well-reasoned.

3. Award zero marks for answers which are contradictory.

4. In questions asking for a specified number of answers, if a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, if a candidate gives three answers, only the first two answers should be marked.

5. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

6. Chemical equations should be balanced except those in reaction schemes for organic synthesis. For energetics, the chemical equations given should include the correct state symbols of the chemical species involved.

7. In the question paper, questions which assess candidates' communication skills are marked with an asterisk (*). For these questions, the mark for effective communication (1 mark per question) will be awarded if the answer is easily understandable. No effective communication marks will be awarded if an answer contains wrong chemistry concept and/or a lot of irrelevant material.

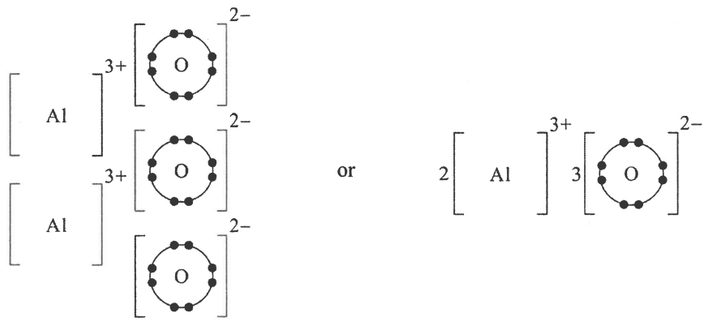
SECTION B

Part I

Marks

1. (a) Aluminium has a giant metallic structure and metallic bonds. 1
When an external force is applied, aluminium ions will shift but without breaking the metallic bonds. The shape of aluminium metal will change but the metal will not break. 1

(b) 1



(c) $Al_2O_3(s)$ is impermeable / impervious to water and oxygen / air, thus the corrosion resistance of aluminium can be enhanced. 1

(d) anodisation 1

2. (a) water / H_2O 1

(b) The crystals change from blue to white / pale blue. 1

(c) Moisten a clean nichrome / platinum wire with concentrated hydrochloric acid and dip the nichrome wire into a small amount of hydrated copper(II) sulphate crystals. 1
Heat the tip of the nichrome wire in a non-luminous flame and observe the flame colour produced. 1

Marks

3. (a) (i) $3Mg(s) + N_2(g) \rightarrow Mg_3N_2(s)$ 1

(ii) No. of moles of Mg $= \frac{1.85}{24.3} = 0.0761$ 3

No. of moles of $N_2 = \frac{2.10}{(14.0 \times 2)} = 0.0750$

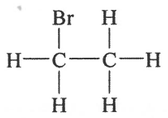
Mole ratio of Mg to $N_2 = 3 : 1$
/ 0.0750 mol of N_2 reacts completely with 0.2250 mol of Mg
/ 0.0761 mol of Mg reacts completely with 0.0254 mol of N_2.
Therefore, N_2 is in excess / Mg is limiting.

The theoretical mass of Mg_3N_2 that can be formed
$= 0.0761 \times \frac{1}{3} \times (24.3 \times 3 + 14.0 \times 2)$
$= 2.56$ g

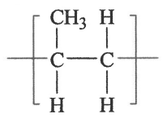
(b) (i) Magnesium burns with a bright white light. 1

(ii) More energy is required to break the stronger triple bond in a N_2 molecule than to break the double bond in an O_2 molecule. Therefore, nitrogen is less reactive than oxygen. 1

4. (a) C_nH_{2n+2} 1

(b) (i)  1

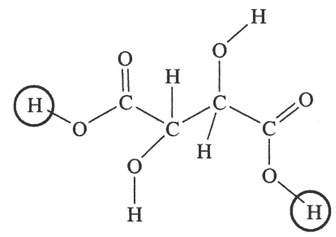
Bromine (dissolved in an organic solvent) changes from orange / brown to colourless / decolourises slowly. 1

(ii)  1

addition polymerisation 1

(c) Polar ice caps melt. / Flooding happens in the low-lying areas. / Sea level rises. / There are abnormal climate changes in different regions of the world. / There is extreme weather in many places. / Rainfall pattern changes. 1

Marks

5. (a)  1

(b) (i) from colourless to pale pink 1

(ii) No. of moles of NaOH used $= \frac{0.0105 \times 16.80}{1000} = 1.764 \times 10^{-4}$ 3

No. of moles of tartaric acid in the white wine sample
$= 1.764 \times 10^{-4} \times \frac{1}{2} \times \frac{100.0}{25.0}$
$= 3.528 \times 10^{-4}$

Concentration of tartaric acid in the white wine sample
$= 3.528 \times 10^{-4} \times 150.0 \times \frac{1000}{25.0}$
$= 2.12$ g dm^{-3}

(iii) To measure the change in temperature of the reaction mixture during titration. 1
Plot a graph of temperature against volume of NaOH(aq). In the graph, there is a maximum temperature of the reaction mixture at the end point. 1

6. (a) To provide mobile ions. / To increase electrical conductivity. 1

(b) The solution around electrode **Y** changes from green to blue. 1
Because the position of $H^+(aq)$ ion is lower than that of $Na^+(aq)$ ion in the electrochemical series / $H^+(aq)$ ion is a stronger oxidising agent than $Na^+(aq)$ ion, $H^+(aq)$ ions are preferentially discharged to form hydrogen gas. 1
During electrolysis, water molecules ionise continuously to replace the discharged $H^+(aq)$ ions. 1
This causes $OH^-(aq)$ ions to accumulate, therefore the concentration of $OH^-(aq)$ ions is higher than that of $H^+(aq)$ ions and the solution around electrode **Y** becomes alkaline.

(c) Electrode **X** becomes thinner. 1
$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ 1

(d) $Zn^{2+}(aq) + 2OH^-(aq) \rightarrow Zn(OH)_2(s)$ 1

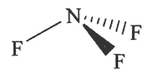
Marks

7. (a) (i) (1)

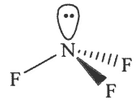


1

(2)



or



1

(ii) BF_3 is a non-polar molecule, because the polarities of the B–F bonds cancel out each other. 1
NF_3 is a polar molecule, because the polarities of the N–F bonds do not cancel out each other. 1

(b) **A > C > B** 1
The viscosity of a substance depends on the ease of movement between particles, which in turn depends on the strength of the attraction between particles. 1
The number of hydrogen bonds formed per molecule on average :
$HOCH_2CH(OH)CH_2OH$ > $CH_3CH(OH)CH_2OH$ > $CH_3CH_2CH_2OH$

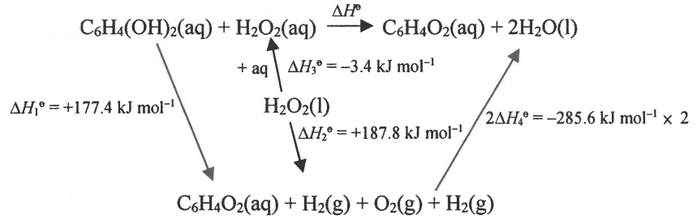
8. (a) (i) Standard enthalpy change of formation of $H_2O_2(l)$ is the enthalpy change when one mole of $H_2O_2(l)$ is formed from its constituent elements ($H_2(g)$ and $O_2(g)$) in their standard states under standard conditions / 25 °C and 1 atm. 1

(ii) Water is formed from the reaction between hydrogen and oxygen. 1

(iii) (1) • one H–H bond and one O=O bond 1
• two H–O bonds and one O–O bond

(2) The total energy absorbed in the bond breaking processes is less than the total energy released in the bond forming processes. 1

(b) (i) 2



(ii) The standard enthalpy change of Reaction (I) (ΔH°) 2
$= \Delta H_1^\circ + \Delta H_2^\circ - \Delta H_3^\circ + 2\Delta H_4^\circ$
$= 177.4 + 187.8 - (-3.4) + 2 \times (-285.6)$
$= -202.6$ kJ mol^{-1}

Marks

9. Chemical knowledge 5

Showing the reaction occurring at each electrode :
- Electrode **X** is the anode / the negative electrode, $H_2(g)$ is the reducing agent / undergoes oxidation / loses electrons at electrode **X**.
- Electrode **Y** is the cathode / the positive electrode, $O_2(g)$ is the oxidising agent / undergoes reduction / gains electrons at electrode **Y**.

Describing the function of the electrolyte :
- The concentrated KOH(aq) provides mobile ions / increases the electrical conductivity.

Describing the functions of other parts of the fuel cell :
Any TWO of the following :
- The energy conversion is from chemical energy to electrical energy.
- Platinum acts as a catalyst / an inert electrode / can increase the reaction rate. / The porous electrodes can increase the surface area of the electrodes to speed up the reactions.
- The continuous supply of hydrogen and oxygen can allow the fuel cell to work continuously. / The unreacted gases (hydrogen and oxygen) are recycled to save chemicals.

Communication mark 1

Part II

Marks

10. (a) Mass of KO_2 needed 2

$$= \frac{140.0}{24.0} \times 2 \times (39.1 + 16.0 \times 2) \times \frac{1}{1000}$$

$= 0.8295$ kg

(b) This reaction can provide oxygen gas for astronauts in spacecrafts. / The products (potassium carbonate and oxygen) from this reaction are non-toxic. 1

(c) From the graph, the slope of the tangent at point **A** is greater than that at point **B**. Therefore, the instantaneous rate of the reaction at point **A** is higher than that at point **B**. 1

11. (a) (i) $[Fe^{3+}(aq)]_{\text{initial after mixing}} = \frac{0.0165 \times 10.0 \times 2}{(10.0 + 15.0)} = 0.0132 \text{ mol dm}^{-3}$ 1

$[Fe^{3+}]_{eqm} = 0.0132 - 0.00912 = 0.00408 \text{ mol dm}^{-3}$

(ii) $[SCN^-(aq)]_{\text{initial after mixing}} = \frac{0.0255 \times 15.0}{(10.0 + 15.0)} = 0.0153 \text{ mol dm}^{-3}$ 2

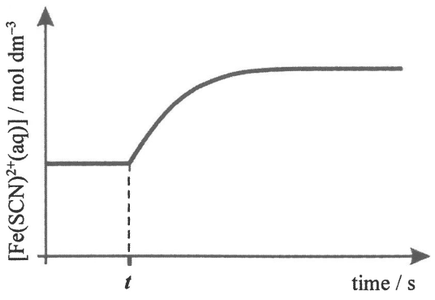
$[SCN^-(aq)]_{eqm} = 0.0153 - 0.00912 = 0.00618 \text{ mol dm}^{-3}$

$$K_c = \frac{[Fe^{3+}(aq)]\,[SCN^-(aq)]}{[Fe(SCN)^{2+}(aq)]}$$

$$= \frac{0.00408 \times 0.00618}{0.00912}$$

$= 0.00276 \text{ mol dm}^{-3}$

(b) 1

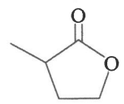


(c) As the temperature decreases, the colour of the solution becomes deeper red. This implies that more $Fe(SCN)^{2+}(aq)$ is formed, and the equilibrium position shifts to the left / reactant side. 1

A decrease in temperature favours exothermic reaction. Therefore, the forward reaction is endothermic. 1

Marks

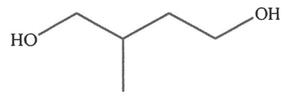
12. (a) (i) 1



(ii) Yes, because the carboxyl group (–COOH) reacts with the hydroxyl group (–OH) to join up the two ends of molecule **A**, and there is an elimination of a small molecule / H_2O. 1

(b) $Cr_2O_7^{2-}/H^+$, heat 1

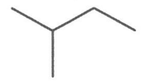
(c) 1



(d) (i) Al_2O_3, heat or conc. H_2SO_4, heat 1

(ii) methylbuta-1,3-diene 1

(e) (i) 1



(ii) Pt / Ni / Pd 1

13. (a) Cl_2O 1

It reacts with water to form hypochlorous acid / HOCl(aq) which is acidic. 1

(b) (i) electroplating / chromium-plating / forming stainless steel 1

(ii) Chromium forms coloured ions: $Cr_2O_7^{2-}(aq)$ ions exhibit orange / $CrO_4^{2-}(aq)$ ions exhibit yellow. 1

14. Chemical knowledge 5

Mentioning the different structures of the two compounds:

- CH_3CH_2COOH (propanoic acid) and
  CH_3COOCH_3 (methyl ethanoate) / $HCOOCH_2CH_3$ (ethyl methanoate)

Explaining the difference in their boiling points:

- Propanoic acid molecules are held by hydrogen bonds, while methyl ethanoate (or ethyl methanoate) molecules are held by van der Waals' forces.
- The intermolecular forces (hydrogen bonds) in propanoic acid are stronger, therefore the boiling point of propanoic acid is higher. /
  The intermolecular forces (van der Waals' forces) in methyl ethanoate (or ethyl methanoate) are weaker, therefore the boiling point of methyl ethanoate (or ethyl methanoate) is lower.

Describing their chemical reactions with NaOH(aq):

- The reaction between propanoic acid and NaOH(aq) is neutralisation, a salt and water are formed.
- The reaction between methyl ethanoate (or ethyl methanoate) and NaOH(aq) under heating is alkaline hydrolysis, a salt of alkanoic acid and an alcohol are formed.

Communication mark 1

Marks

1. (a) (i) $2NaCl(aq) + 2H_2O(l) \rightarrow Cl_2(g) + H_2(g) + 2NaOH(aq)$ 1

(ii) Add a catalyst. 1
The reaction can proceed through an alternative reaction pathway with a lower activation energy, E_a(II). 1

(iii) Any TWO of the following: 2
- Method 1 has a higher atom economy than Method 2.
- Non-toxic H_2O is produced in Method 1 while toxic SO_2 / corrosive HCl is produced in Method 2.
- Method 1 uses a catalyst but Method 2 does not.

(b) (i) To remove impurities which may poison the catalyst. 1

(ii) It is more difficult and expensive to obtain pure O_2(g). / It is easier and cheaper to obtain air to provide O_2(g) for the reaction. 1

(iii) A catalyst in porous form can provide a larger surface area that increases the effectiveness of the catalyst. 1

(iv) (1) Increasing temperature can increase the reaction rate, but the forward reaction is exothermic, so the higher temperature can shift the equilibrium position to the left / reactant side and thus decrease the yield of sulphur trioxide / sulphuric acid. Therefore, 1 000 °C is NOT chosen. 1

(2) Increasing pressure can increase the reaction rate and increase the yield by shifting the equilibrium position to the right, but the yield of sulphur trioxide at equilibrium is already very high (98%). Increasing the pressure can only lead to a very small increase in the yield. 1
The construction cost and maintenance cost of a plant that can withstand high pressure are high. / High pressure may cause explosions or safety concerns. Therefore, 200 atm is NOT chosen. 1

(v) $2NH_3(aq) + H_2SO_4(aq) \rightarrow (NH_4)_2SO_4(aq)$ 1

Marks

1. (c) (i) Any TWO of the following: 2
- To ensure the initial concentration of NaOH(aq) is the only variable.
- To keep the concentration of **Y**(aq) constant.
- To ensure fair comparisons between the trials.

(ii) $$\frac{\text{initial rate 1}}{\text{initial rate 2}} = \frac{8.72 \times 10^{-3}}{4.36 \times 10^{-3}} = \frac{k[\mathbf{Y}(aq)]^m \left[2.0 \times \frac{8.0}{(8.0+2.0)}\right]^n}{k[\mathbf{Y}(aq)]^m \left[2.0 \times \frac{4.0}{(4.0+4.0+2.0)}\right]^n}$$ 1

Therefore, $n = 1$ / The order of reaction with respect to OH^-(aq) is 1.

Or,
Comparing Trial 1 with Trial 2, the concentration of OH^-(aq) has halved and the initial rate has also halved.
Therefore, $n = 1$ / The order of reaction with respect to OH^-(aq) is 1.

(iii) Comparing Trial 1 with Trial 3, the concentration of OH^-(aq) has decreased by a factor of $\frac{3}{8}$.
$\therefore$ The initial rate of the disappearance of **Y**(aq) $= 8.72 \times 10^{-3} \times \frac{3}{8}$ 1
$= 3.27 \times 10^{-3}$ mol dm^{-3} s^{-1}

Or,
Comparing Trial 2 with Trial 3, the concentration of OH^-(aq) has decreased by a factor of $\frac{3}{4}$.
$\therefore$ The initial rate of the disappearance of **Y**(aq) $= 4.36 \times 10^{-3} \times \frac{3}{4}$
$= 3.27 \times 10^{-3}$ mol dm^{-3} s^{-1}

(iv) (1) Perform another three trials of the experiment to determine the order of reaction with respect to **Y**(aq). In each trial, different volumes of 0.012 M **Y**(aq) and H_2O(l) were mixed with a fixed volume of 2.0 M NaOH(aq) to measure the initial rate of disappearance of **Y**(aq). The total volume of the reaction mixture was kept constant in each trial. 1

Plot a graph of initial rate of disappearance of **Y**(aq) against initial concentration of **Y**(aq). 1
A straight line passing through the origin is obtained, which shows a directly proportional relationship between rate and concentration. 1

Or, Plot a graph of *log* (initial rate of disappearance of **Y**(aq)) against *log* (initial concentration of **Y**(aq)).
The slope of the straight line obtained is equal to 1.

(2) mol^{-1} dm^3 s^{-1} 1

Marks

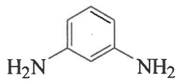
2. (a) (i) The size of these plastic particles ranges from 50 nm to 90 nm. The size of nanoparticles is typically between 1 nm and 100 nm. Therefore, these plastic particles are nanoparticles. 1

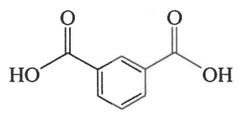
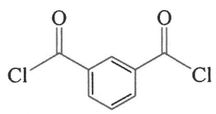
(ii) (1) Yes, **Z** can. Because the molecule of **Z** is chiral. 1

(2) D / milky fluid 1

(iii) Water molecules can form hydrogen bonds with hydroxyl groups / –OH groups in the cellulose polymer chains. Therefore, cotton absorbs water easily. 1

Cellulose consists of many hydroxyl groups / –OH groups that can form hydrogen bonds between adjacent cellulose polymer chains. However, the water molecules cannot break all the hydrogen bonds between the long cellulose polymer chains. Therefore, cotton is insoluble in water. 1

(b) (i)  1

 or  1

(ii) The atom economy is not 100%, because the reaction does not convert all the reactant atoms to the desired product (polymer **X**) / there is by-product (H_2O / HCl) formed in the reaction / the reaction involves the elimination of small molecules (H_2O / HCl). 1

(iii) There are many strong intermolecular hydrogen bonds between the polymer chains of **X**. 1

X has many benzene rings which restrict the movement between the polymer chains. 1

(iv) high heat resistance / low density 1

(v) Because the amide groups in **X** can undergo acid hydrolysis. 1

(vi) Wool is a renewable resource, but **X** comes from non-renewable resources (e.g. petroleum). 1

(c) (i) (1) hexagonal close-packed structure 1

(2) 12 1

(ii) Number of titanium atoms in the unit cell $= 8 \times \frac{1}{8} + 1 = 2$ 1

(iii) (1) The different sizes of metal **Y** atoms and titanium atoms distort the regular packing of atoms in pure titanium. This greatly hinders the sliding of the layers of atoms over each other, so this titanium alloy is harder than pure titanium. 1

(2) biocompatible / corrosion resistant 1

(iv) **B** 1

The higher the coordination number in the metallic crystal structure, the more closely packed are the atoms, so there is less empty space in the structure. The coordination number of each titanium atom in crystal structure **B** is the smallest among the three crystal structures. 1

Marks

3. (a) (i) fractional distillation 1

(ii) Pass the gas to KI(aq). 1

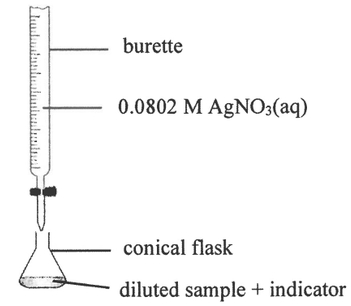
The solution changes from colourless to brown. 1

Or, Pass the gas to KBr(aq).

The solution changes from colourless to reddish brown.

(iii) (1) $R_f = \frac{3.8}{(13.0-2.0-1.0)} = 0.38$ 1

(2) Paracetamol adsorbs less strongly onto the stationary phase than caffeine. 1

(b) (i) 

2

(ii) $(7.80 + 7.75 + 7.85) \div 3 = 7.80\ cm^3$ 1

(iii) Concentration of $Cl^-(aq)$ ions in the impure water sample 3

$= \frac{7.80}{1000} \times 0.0802 \times \frac{1000}{10.0} \times \frac{250.0}{25.0} \times 35.5 \times \frac{1}{1000}$

$= 2.221 \times 10^{-2}\ g\ cm^{-3}$

(iv) The calculated concentration of the $Cl^-(aq)$ ions would be higher than the actual value / overestimated. Because $Br^-(aq)$ ions can form precipitate with $Ag^+(aq)$ ions, so a larger volume of $AgNO_3(aq)$ is needed to reach the end point of the titration. 1

Marks

3. (c) (i) 2

$$\text{No. of C atoms} = \frac{100 \times \frac{72.0}{100}}{12.0} = 6$$

$$\text{No. of H atoms} = \frac{100 \times \frac{12.0}{100}}{1.0} = 12$$

$$\text{No. of O atoms} = \frac{100 \times \frac{(100-72.0-12.0)}{100}}{16.0} = 1$$

The molecular formula of **Q** is $C_6H_{12}O$.

Or,

$$\text{Mole ratio of C : H : O} = \frac{72.0}{12.0} : \frac{12.0}{1.0} : \frac{(100-72.0-12.0)}{16.0}$$

$$= 6 : 12 : 1$$

Empirical formula is $C_6H_{12}O$.

Let the molecular formula be $(C_6H_{12}O)_n$.
$n\,(12.0 \times 6 + 1.0 \times 12 + 16.0) = 100$
$n = 1$

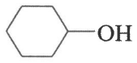
The molecular formula of **Q** is $C_6H_{12}O$.

(ii) (1) When molecules absorb infra-red radiation of certain characteristic frequencies corresponding to the vibrational frequencies of a specific bond in the molecule, this gives rise to an absorption peak in an infra-red spectrum. 1

(2) The infra-red spectrum of **Q** shows a strong absorption peak at around 3 350 cm^{-1} corresponding to O–H group of alcohol. 1
The infra-red spectrum of **R** shows a strong absorption peak at around 1 700 cm^{-1} corresponding to C=O group. 1

(iii) The infra-red spectrum of **R** does not show a strong absorption peak in the wavenumber range of 2 500 cm^{-1} – 3 300 cm^{-1} corresponding to O–H bond of carboxylic acid, so **R** is not a carboxylic acid. 1
Q is oxidised to form **R**, and **R** is not a carboxylic acid, so it can be deduced that **Q** is a secondary alcohol and **R** is a ketone. 1
Or, **Q** is optically inactive, which implies that **Q** does not have any chiral carbon centre.

The possible structure of **Q** is 1



Candidates' Performance

Paper 1

Paper 1 consisted of two sections: Section A (multiple-choice questions) and Section B (conventional questions). Sections A and B each comprised two parts: Part I and Part II. Part I contained questions mainly on Topics I to VIII of the curriculum, while Part II contained questions mainly on Topics IX to XII. All questions in both sections were compulsory.

Section A (multiple-choice questions)

This section consisted of 36 multiple-choice questions. The mean score was 23.4. The performance of candidates was generally good. Some misconceptions were revealed by candidates in their performance on the following items:

1. For Q.21, about a third of the candidates correctly chose answer D. It should be noted that all the three reactions shown in this item are redox reactions. The reaction in (1) involves the addition of hydrogen to an alkene. The two carbon atoms in the carbon-carbon double bond are reduced while hydrogen atoms in the H_2 molecule are oxidised. Many candidates wrongly thought that the reaction in (1) involves reduction only as there is a gain of hydrogen in $CH_3CH{=}CHCH_3$. This may explain why nearly half of the candidates wrongly chose option C as the answer.

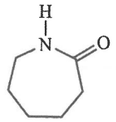
Q.21 Which of the following reactions involve oxidation ?

(1) $CH_3CH{=}CHCH_3 + H_2 \rightarrow CH_3CH_2CH_2CH_3$
(2) $3Br_2 + 6NaOH \rightarrow 5NaBr + NaBrO_3 + 3H_2O$
(3) $2Al + 2NaOH + 6H_2O \rightarrow 2NaAl(OH)_4 + 3H_2$

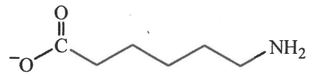
A. (1) and (2) only (5%)
B. (1) and (3) only (9%)
C. (2) and (3) only (49%)
D. * (1), (2) and (3) (37%)

2. For Q.33, nearly half of the candidates correctly chose answer A. In this item, the reaction between the given organic compound and excess NaOH(aq) is an alkaline hydrolysis, not a condensation reaction. The amide bond in the ring structure of the organic compound is cleaved to form a $-COO^-$ ion at one end of the carbon chain and a $-NH_2$ group at the other end. Therefore, statement (1) is correct. About a quarter of the candidates wrongly chose option B as the answer. They might wrongly think that the organic compound was an unsubstituted amide and thus ammonia would be formed in the reaction.

Q.33 The following organic compound is heated with excess NaOH(aq).



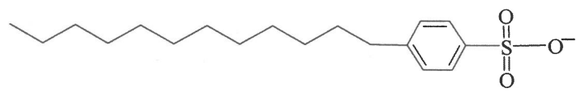
Which of the following statements about this reaction is / are correct ?

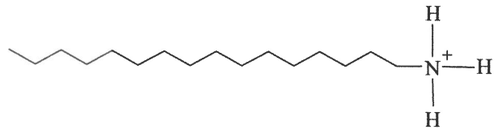
(1)  is formed.
(2) Ammonia is formed.
(3) It is a condensation reaction.

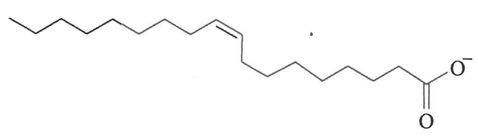
A. *	(1) only	(46%)
B.	(2) only	(25%)
C.	(1) and (3) only	(16%)
D.	(2) and (3) only	(13%)

3. For Q.35, the species that can emulsify an oil-water mixture should be detergents. The structure of a detergent should include a long hydrophobic tail and a hydrophilic head. About half of the candidates wrongly chose option B as the answer. They might not recognise that there is an ionic head which is hydrophilic in the species shown in (2).

Q.35 Which of the following species can emulsify an oil-water mixture ?

(1) 

(2) 

(3) 

A.	(1) and (2) only	(13%)
B.	(1) and (3) only	(52%)
C.	(2) and (3) only	(8%)
D. *	(1), (2) and (3)	(27%)

Section B (conventional questions)

Question Number	Performance in General
1	The performance of candidates was good. In part (a), about half of the candidates were able to state that aluminium has a giant metallic structure and metallic bonds exist between atoms. Just under half of the candidates were able to give a complete explanation for aluminium being malleable. Many candidates did not mention the changes at the atomic level when an external force is applied. A small number of the candidates wrongly stated that the bonding in solid aluminium is an ionic bond. In part (b), about three quarters of the candidates were able to correctly draw the electron diagram for aluminium oxide. A small number of the candidates wrongly considered aluminium oxide to be a covalent compound and drew an incorrect electron diagram. In part (c), about three quarters of the candidates were able to explain why the aluminium oxide layer can enhance the corrosion resistance of aluminium. In part (d), about two thirds of the candidates were able to correctly write the name 'anodisation'. Some candidates mixed up the process with 'sacrificial protection' and galvanisation'.
2	The performance of candidates was fair. In part (a), a very high percentage of candidates were able to state that the colourless liquid collected is water. In part (b), about half of the candidates were able to write the expected colour change. In part (c), about a quarter of the candidates were able to correctly describe how the flame test is carried out in a school laboratory. Some candidates wrongly suggested using sulphuric acid or nitric acid to conduct the test. A small number of candidates failed to mention the use of platinum wire or nichrome wire, as well as the necessity of using a non-luminous flame to heat the sample.
3	The performance of candidates was satisfactory. In part (a)(i), a very high percentage of candidates were able to correctly give the chemical equation for the reaction. In part (a)(ii), about half of the candidates were able to give the correct steps and answer in the calculation. Some candidates did not clearly show that magnesium is the limiting reactant. In addition, some candidates wrongly calculated the number of moles of N atoms instead of N_2 molecules. A small number of candidates failed to realise that the mole ratio of Mg to N_2 in the reaction is 3:1, resulting in incorrect answers. In part (b)(i), about half of the candidates were able to precisely state that magnesium burns with bright white light. In part (b)(ii), about a third of the candidates were able to explain that magnesium nitride is not the main product in terms of bonding. A significant number of candidates failed to mention the double bonds in O_2 molecules and the triple bonds in N_2 molecules. Moreover, they wrongly suggested that nitrogen is less reactive than oxygen because there are stronger van der Waals' forces between N_2 molecules than between O_2 molecules.

Question Number	Performance in General
4	The performance of candidates was good. In part (a), a very high percentage of candidates were able to state the correct general formula for the homologous series. In part (b)(i), about two thirds of the candidates were able to give the correct structural formula for the main organic product of the reaction. Some candidates failed to read the question carefully, not noticing that a large excess of ethane is used, and therefore incorrectly suggesting polysubstituted products. A small number of candidates failed to precisely describe the expected observation when ethane reacts with bromine (dissolved in an organic solvent), and a few candidates wrongly stated that the colour changes from colourless to brown. In part (b)(ii), about half of the candidates were able to correctly draw the repeating unit of the polymer formed from propene and name the type of polymerisation. Some candidates wrongly drew the structure of the monomer or added 'n' in the drawing. In part (c), about two thirds of the candidates were able to state an environmental problem caused by global warming. Some candidates failed to present their ideas clearly. For example, they gave 'rising temperatures' without mentioning the relevant consequence and 'melting ice' without mentioning the polar ice caps. Moreover, some candidates provided irrelevant environmental problems, such as acid rain.
5	The performance of candidates was satisfactory. In part (a), about two thirds of the candidates were able to identify the ionisable hydrogen atoms in the structure of tartaric acid. However, a small number of candidates did not read the question carefully and wrongly circled the carboxyl groups. In part (b)(i), a very high proportion of the candidates were able to state the correct colour change of phenolphthalein at the end of the titration. In part (b)(ii), fewer than half of the candidates were able to give the correct steps and answer in the calculation. Some candidates did not recognise that tartaric acid reacts with NaOH(aq) in a mole ratio of 1:2. Common mistakes included 'failed to calculate the number of moles of acid before dilution' and 'confused how to convert the unit from mol dm^{-3} to g dm^{-3}'. In part (b)(iii), a small number of candidates were able to recognise that the intense red color of red wine masked the colour of the indicator used in the titration. Therefore, they correctly suggested other methods, such as thermometric titration, and clearly described how to determine the end point. About a quarter of the candidates correctly suggested using a pH meter to determine the end point but incorrectly stated that the pH is 7 at the end point in the titration. Some candidates merely suggested that the red pigment should be removed before titration without mentioning a correct modification for the experiment.

Question Number	Performance in General
6	The performance of candidates was satisfactory. In part (a), a high proportion of the candidates were able to state that $Na_2SO_4(aq)$ contains mobile ions, therefore the solution can conduct electricity. However, some candidates gave vague and incomplete answers such as 'it is used to complete the circuit' or 'it contains ions'. In addition, a few candidates wrongly suggested that 'the solution allows electrons to pass through'. In part (b), only a small number of candidates were able to give the correct colour change and a complete explanation. Although a large proportion of the candidates were able to state that the solution turns blue, many of them did not mention the initial colour of the solution. In their explanation of the colour change, many candidates were able to mention that $H^+(aq)$ ions are discharged at the electrode, but they failed to explain explicitly why $H^+(aq)$ ions are more preferentially discharged than $Na^+(aq)$ ions by comparing the relative oxidising power of $H^+(aq)$ ions and $Na^+(aq)$ ions. Moreover, a significant number of candidates were unable to point out that the accumulation of excess $OH^-(aq)$ ions around the electrode is the reason why the solution turns blue. In part (c), about two thirds of the candidates were able to state the correct change at the electrode and write a correct half equation. However, a small number of candidates gave wrong answers such as 'colourless gas is evolved' or 'a white precipitate is formed'. Moreover, a small number of candidates wrote a wrong half equation such as $Zn(s) + 2e^- \rightarrow Zn^{2+}(aq)$ or $Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$. In part (d), fewer than half of the candidates were able to give the correct ionic equation for the formation of $Zn(OH)_2(s)$. Some candidates wrongly suggested that the white precipitate is $ZnSO_4$. A very small number of candidates gave an unbalanced ionic equation or wrongly wrote the chemical formula of zinc hydroxide as ZnOH.
7	The performance of candidates was good. In part (a)(i)(1), a very high percentage of the candidates were able to draw the correct three-dimensional structure of BF_3. In part (a)(i)(2), about three quarters of the candidates were able to draw the correct three-dimensional structure of NF_3. However, some candidates were not able to use correct notations to represent the N-F bonds 'in front of' and 'behind' the plane respectively and they failed to illustrate the trigonal pyramidal shape of the NF_3 molecule. Moreover, a small number of candidates wrongly drew two lone pairs of electrons on the nitrogen atom of the NF_3 molecule or wrongly suggested that the NF_3 molecule is trigonal planar or tetrahedral. In part (a)(ii), just under half of the candidates were able to give the correct polarity of the two given molecules with complete explanation. About a third of the candidates failed to explicitly state the reasons for the molecules being polar or non-polar. They gave wrong answers such as 'the polar bonds can/cannot cancel each other' instead of correctly writing 'the bond polarities can/cannot cancel each other'. In part (b), a high percentage of the candidates were able to arrange the three liquids in descending order of viscosity. However, only a third of the candidates were able to give the correct explanation in terms of the number of hydrogen bonds per molecule formed by each liquid. Some candidates wrongly give explanations in terms of the difference in molecular size among the three alcohols, and hence wrongly suggested a reason by referring to the difference in strength of van der Waals' forces. A minority of candidates only mentioned the difference in the number of hydroxyl groups among the molecules of the three alcohols without giving further explanation in terms of the number of hydrogen bonds being formed. A small number of candidates wrongly explained the viscosity of the liquids in terms of the interaction between the alcohol molecules and water molecules.

Question Number	Performance in General
8	The performance of candidates was poor. In part (a)(i), about a quarter of the candidates were able to state the meaning of the standard enthalpy change of formation of $H_2O_2(l)$. Many candidates gave incomplete answers with missing keywords such as '… the enthalpy change of…', 'one mole of…', 'from its constituent elements', 'in their standard states' and 'under standard conditions'. In addition, a small number of candidates failed to give the correct physical states of the substances involved. In part (a)(ii), about a third of the candidates were able to explain that the enthalpy change cannot be determined directly by experiments due to the formation of water. A small number of candidates gave vague answers such as 'side reaction occurs', 'side product is formed' or '$H_2O_2(l)$ cannot be formed easily'. A minority of candidates gave clearly mistaken answers, including 'the reaction is very slow', 'the reaction is explosive', 'heat loss to the surrounding' or 'the reaction is not performed under standard conditions'. In part (a)(iii)(1), about a third of the candidates were able to state the correct type and number of bonds broken and formed. About two thirds of the candidates were able to correctly state the type of bonds broken in H_2 and O_2, but missed the type of bonds formed in H_2O_2. Some candidates wrongly suggested that the bond between the two oxygen atoms in H_2O_2 is a triple bond or a dative covalent bond. Moreover, about a quarter of the candidates only gave the type of bonds broken and formed without mentioning the corresponding number of bonds, or only gave the number of bonds broken and formed without stating the type of bonds. In part (a)(iii)(2), about half of the candidates were able to explain why the enthalpy change is a negative value. Nevertheless, a small number of candidates wrongly suggested that the reaction is exothermic because the number of bonds formed is greater than the number of bonds broken, or wrongly thought that bond breaking releases energy while bond forming absorbs energy. Furthermore, some candidates gave vague answers such as 'the energy involved in bond breaking is less than the energy involved in bond forming' without explicitly stating whether the energy is absorbed or released. In part (b)(i), only a small number of candidates were able to draw an enthalpy change cycle with correct labels and balanced equations. Some candidates drew an enthalpy change cycle with unbalanced equations, wrong direction of arrows, incorrect labels of ΔH values or missing state symbols for the substances involved. In part (b)(ii), just under half of the candidates were able to give the correct steps and answer in the calculation.
9	The performance of candidates was poor. A very small number of candidates were able to use the information provided in the diagram to elaborate on the working principle of the fuel cell, and give a correct, complete and well-presented answer. A high proportion of candidates were able to mention that concentrated KOH(aq) is the electrolyte, $H_2(g)$ is the fuel and $O_2(g)$ is the oxidant. Nevertheless, a small number of candidates wrongly thought that KOH(aq) is a catalyst. About two thirds of the candidates were able to give a correct half equation for the reaction at the cathode, but some candidates wrongly gave a half equation for the reaction at the anode. They wrongly suggested that $H_2(g)$ is oxidised to $H^+(aq)$ ions in the alkaline medium. Moreover, nearly half of the candidates failed to mention the direction of electron flow in the external circuit of the fuel cell, or did not explicitly state that electrode **X** is the anode (negative pole) while electrode **Y** is the cathode (positive pole). About a quarter of the candidates were able to state that the platinum can catalyse the reactions at the electrodes. Only a small number of candidates were able to mention other important points, including 'the continuous supply of $H_2(g)$ and $O_2(g)$ can ensure the fuel cell works continuously', 'the unreacted $H_2(g)$ and $O_2(g)$ can be recycled to save chemicals' and 'the fuel cell converts chemical energy to electrical energy'. Some candidates gave answers that were irrelevant to the working principle, for example, 'the only product in the fuel cell is water which is non-polluting' or 'this fuel cell is a primary cell'.

Question Number	Performance in General
10	The performance of candidates was satisfactory. In part (a), just under half of the candidates were able to give the correct steps and answer in the calculation. Some candidates did not read the question carefully and wrongly gave the answer with unit 'g', instead of 'kg' as required by the question. In part (b), about two thirds of the candidates were able to give the correct answer. A small number of candidates wrongly stated that the method mentioned in the question was effective in reducing greenhouse gases. They failed to recognise that such an answer is irrelevant in the context of spacecraft. In part (c), about two thirds of the candidates were able to state that the instantaneous rate of the reaction is higher at point **A** and consider the slope of the tangent at point **A** to give the correct deduction. A small number of candidates wrongly stated that the instantaneous rate is higher at point **B** because the pressure of the system is higher at that instant. A few candidates did not refer to the information given in the graph and merely stated that the reaction rate at point **A** is higher due to the higher concentrations of the reactants at that moment.
11	The performance of candidates was fair. In part (a)(i), about a quarter of the candidates were able to correctly calculate the concentration of $Fe^{3+}(aq)$ in the equilibrium mixture. Some candidates failed to account for the final volume of the equilibrium mixture in their calculations, while some candidates wrongly gave the initial concentration of $Fe^{3+}(aq)$ in the system as the answer. Some candidates failed to realise that there are two Fe^{3+} ions in each formula unit of $Fe_2(SO_4)_3$. In part (a)(ii), about a quarter of the candidates were able to give the correct steps and answer in the calculation. Although more than half of the candidates were able to give the correct equilibrium expression for the system, many candidates failed to find out the correct concentrations of the reacting species needed to perform the calculation. A small number of candidates wrongly gave the equilibrium expression as $\frac{[Fe(SCN)^{2+}(aq)]}{[Fe^{3+}(aq)]\,[SCN^{-}(aq)]}$. In part (b), just under half of the candidates were able to draw a correct sketch showing the variation in the concentration of $Fe(SCN)^{2+}(aq)$ with time. Some candidates incorrectly indicated a sudden change in the concentration of $Fe(SCN)^{2+}(aq)$ at time t, which does not align with the expected response of the system. In part (c), about a quarter of the candidates were able to give a complete and correct answer. About half of the candidates were able to correctly state that the equilibrium position shifts to the left as the temperature decreases, based on the information provided. However, some candidates did not explicitly state the reason required to deduce that the forward reaction is an endothermic reaction, for example 'a decrease in temperature favours exothermic reactions'. A small number of candidates wrongly stated that the rate of backward reaction increases as the temperature decreases.

Question Number	Performance in General
12	The performance of candidates was satisfactory. In part (a)(i), about a third of the candidates were able to give the correct structural formula of compound **B**. Many candidates failed to recognise that compound **B** is a cyclic compound and wrongly gave the structure of an acyclic compound, while suggesting that compound **A** undergoes dehydration at the hydroxyl-bearing end to form a carbon-carbon double bond. A small number of candidates gave a structural formula for a compound with an incorrect number of carbon atoms. In part (a)(ii), about half of the candidates were able to give the correct answer. A small number of candidates failed to identify the type of reaction involved and wrongly stated that no water molecule is eliminated during the reaction. In part (b), about three quarters of the candidates were able to identify that the reaction involved in the conversion is oxidation and suggested an appropriate oxidising agent with the corresponding reaction conditions. However, a small number of candidates failed to mention that the oxidising agent $K_2Cr_2O_7$ must be used under acidic conditions. In part (c), about half of the candidates were able to give the correct structural formula of compound **D**. Many candidates correctly identified the reaction as a reduction. However, a small number of candidates drew a structural formula with an incorrect carbon chain length or omitted hydrogen atoms from some of the carbon atoms. In part (d)(i), about two thirds of the candidates were able to identify the reaction as the dehydration of alcohol and suggested an appropriate reagent with the corresponding reaction conditions. However, a small number of candidates failed to explicitly state that the sulphuric acid used must be 'concentrated', or failed to mention that 'heating' is required to carry out the reaction. In part (d)(ii), only about a quarter of the candidates were able to give the correct systematic name of compound **E**. Many of the candidates were not familiar with the conventions for naming dienes. A common error was the omission of the 'a' in the spelling of 'methylbuta-1,3-diene'. Additionally, a small number of candidates gave incorrect numbers indicating the position of the carbon-carbon double bond. In part (e)(i), about three quarters of the candidates were able to give the correct structural formula for compound **F**. A small number of candidates omitted hydrogen atoms from some of the carbon atoms in their drawn structures. In part (e)(ii), about three quarters of the candidates were able to suggest a suitable catalyst to be used for the hydrogenation involved in the conversion of compound **E** to compound **F**.

Question Number	Performance in General
13	The performance of candidates was satisfactory. In part (a), about a third of the candidates were able to give the correct answer with complete explanation. Although about three quarters of the candidates correctly identified Cl_2O as an acidic oxide, many of them failed to state the product formed when Cl_2O reacts with water or wrongly stated that 'HCl and HOCl' are formed. A small number of candidates mistakenly stated that both Cl_2O and SiO_2 form solutions with a pH lower than 7 when added to water. In part (b)(i), about two thirds of the candidates were able to suggest an appropriate method that chromium can be used to prevent iron from rusting. However, some candidates gave incorrect answers, such as 'sacrificial protection' or 'connecting a block of chromium to the iron object'. In part (b)(ii), about a third of the candidates were able to correctly state that chromium forms coloured ions and support their answers with reference to the information provided in the question. A small number of candidates mistakenly stated that the oxidation numbers of chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are different. Some candidates gave incorrect colours of the ions, for example that either CrO_4^{2-} or $Cr_2O_7^{2-}$ is green. A few candidates gave incorrect descriptions of the characteristics of transition metals, such as 'transition metals are coloured'.
14	The performance of candidates was poor. Only a small number of candidates were able to give a correct, complete, and well-presented answer, while fewer than half of the candidates were able to give a partially correct answer. Many candidates were able to identify one of the acyclic compounds with the molecular formula $C_3H_6O_2$ as an alkanoic acid and the other as an ester. However, some candidates were not able to give the exact structures of the two compounds. Upon explaining the difference in boiling points between the two compounds, some candidates incorrectly compared the strength of the intermolecular forces between each compound and water, rather than the interaction between the molecules of the compounds. Upon describing the reaction between CH_3CH_2COOH and NaOH(aq), some candidates omitted water as one of the products. Upon describing the reaction between the ester and NaOH(aq), some candidates wrongly stated that no reaction occurs, while some candidates failed to recognise that the reaction medium is strongly alkaline and mistakenly stated that an alkanoic acid, instead of a salt of alkanoic acid, is the product.

Paper 2

Paper 2 consisted of three sections. Section A contained questions set on Topic XIII 'Industrial Chemistry', Section B on Topic XIV 'Materials Chemistry' and Section C on Topic XV 'Analytical Chemistry'. Candidates were required to attempt all questions in two of the sections.

Question Number	Popularity (%)	Performance in General
Section A: 1	98	The performance of candidates in part (a) was satisfactory. In part (a)(i), fewer than half of the candidates were able to write an overall chemical equation for the electrolysis of brine using a membrane electrolytic cell in the chloroalkali industry. In part (a)(ii), about two thirds of the candidates were able to state that the use of a catalyst would provide an alternate pathway with lower activation energy. Some candidates wrongly suggested the use of a higher temperature. In part (a)(iii), about half of the candidates were able to suggest two reasons to explain why Method 1 can be considered as greener than Method 2. Some candidates did not write the answer in a comparative sense. The performance of candidates in part (b) was satisfactory. In part (b)(i), just under half of the candidates were able to explain why sulphur dioxide and air need to be purified before passing into the reaction chamber. In part (b)(ii), just under half of the candidates were able to explain the use of air instead of pure $O_2(g)$ in terms of cost and availability. In part (b)(iii), about three quarters of the candidates were able to explain why solid catalysts used in industrial processes are usually in porous form instead of in the form of lumps. They were able to mention that there is an increase in the surface area of the catalyst resulting in an increase in its effectiveness. In part (b)(iv)(1), about half of the candidates were able to explain why the operation temperature is not chosen at about 1 000 °C in terms of shifting of equilibrium position. In part (b)(iv)(2), fewer than half of the candidates were able to suggest two reasons to explain why the operation pressure is not chosen at about 200 atm. Some candidates did not recognise that the yield is already very high at atmospheric pressure and hence it is not cost effective to use a high pressure of 200 atm. In part (b)(v), about half of the candidates were able to write a chemical equation for the reaction between sulphuric acid and ammonia. The performance of candidates in part (c) was fair. In part (c)(i), fewer than half of the candidates were able to suggest two reasons to explain why the total volume of the reaction mixture was made the same in each trial. In part (c)(ii), about two thirds of the candidates were able to deduce the order of reaction with respect to $OH^-(aq)$. In part (c)(iii), about two thirds of the candidates were able to deduce the initial rate of disappearance of **Y**(aq) in Trial 3. In part (c)(iv)(1), about a third of the candidates were able to describe an experiment with the use of a suitable graph to confirm that the reaction is first order with respect to **Y**(aq). Some candidates swapped the labels of the x-axis and y-axis. In part (c)(iv)(2), about half of the candidates were able to state the unit of the rate constant.

Question Number	Popularity (%)	Performance in General
Section B: 2	6	The performance of candidates in part (a) was very poor. In part (a)(i), a small number of the candidates were able to convert the unit 'm' to 'nm', and mention that the sizes of the plastic particles are within the range of the typical size of nanoparticles. In part (a)(ii)(1), a small number of the candidates were able to correctly state that **Z** is a chiral compound and thus it can exhibit the cholesteric phase of liquid crystals. In part (a)(ii)(2), about a third of the candidates were able to choose 'milky fluid' as the correct description of the appearance of **Z**. In part (a)(iii), about a quarter of candidates were able to explain why cotton absorbs water. However, only a very small number of the candidates were able to explain why cotton is insoluble in water. The performance of candidates in part (b) was poor. In part (b)(i), about a quarter of the candidates were able to give the correct structures of the two monomers of **X**. Some candidates wrongly gave the monomers of Kevlar as the answer. In part (b)(ii), about a third of the candidates were able to point out that small molecules are eliminated during the formation of **X**, so the atom economy of the reaction is not 100%. In part (b)(iii), a small number of the candidates were able to give correct and complete reasons to explain why **X** has a high tensile strength. In part (b)(iv), about a third of the candidates were able to suggest another property of **X** rendering it suitable for making protective jackets for firemen. In part (b)(v), about a quarter of the candidates were able to suggest that the amide groups will undergo acid hydrolysis when exposed to strong acids. In part (b)(vi), about a quarter of the candidates were able to mention that wool is a renewable resource while **X** comes from non-renewable resources. The performance of candidates in part (c) was poor. In part (c)(i)(1), about a third of the candidates were able to name the type of crystal structure. In part (c)(i)(2), about a third of the candidates were able to give the correct coordination number of a titanium atom. In part (c)(ii), about a third of candidates were able to deduce the number of titanium atoms in the unit cell shown. A small number of candidates only gave the answer without any deduction. In part (c)(iii)(1), about a quarter of the candidates were able to state the difference in sizes between atom **Y** and a titanium atom as the reason for the greater hardness of the alloy. In part (c)(iii)(2), fewer than half of the candidates were able to suggest another property that the alloy should have for it to be suitable for making artificial human joints. In part (c)(iv), fewer than half of the candidates were able to choose structure **B** as the correct answer and only a small number of candidates were able to give a correct explanation. Some candidates gave the wrong coordination number of the metal atom in each crystal structure.

Question Number	Popularity (%)	Performance in General
Section C: 3	96	The performance of candidates in part (a) was fair. In part (a)(i), fewer than half of the candidates were able to suggest a physical method to increase the concentration of an aqueous ethanol solution from 12% to about 50%. Some candidates were unable to spell 'fractional distillation' correctly. In part (a)(ii), fewer than half of the candidates were able to suggest a chemical test for detecting chlorine gas. Some candidates suggested using pH or blue litmus paper to detect chlorine gas but did not mention that the testing paper must be moistened with water. In part (a)(iii)(1), about three quarters of the candidates were able to calculate the R_f value for paracetamol. Some candidates did not show the step for the calculation. In part (a)(iii)(2), fewer than half of the candidates were able to explain the result in the chromatogram based on the principle of chromatography. Some candidates confused the two words 'adsorb' and 'absorb'. The performance of candidates in part (b) was fair. In part (b)(i), about half of the candidates were able to draw the titration set-up. Some candidates used a beaker instead of a conical flask to contain the diluted sample. Some candidates did not know the position of the stopcock in the burette. In part (b)(ii), fewer than half of the candidates were able to give a reasonable average volume of $AgNO_3(aq)$ used. Some candidates did not recognise that the reasonable average volume should be written to 2 decimal places. In part (b)(iii), about half of the candidates were able to calculate the concentration of $Cl^-(aq)$ ions in the impure water sample. In part (b)(iv), about a quarter of the candidates were able to explain that the concentration of $Cl^-(aq)$ ions would be higher than expected in the presence of $Br^-(aq)$ ions. Some candidates were not aware that $Br^-(aq)$ ions would also form a precipitate with $AgNO_3(aq)$. The performance of candidates in part (c) was poor. In part (c)(i), about three quarters of the candidates were able to deduce the molecular formula of **Q**. In part (c)(ii)(1), a very small number of candidates were able to state the principle in producing absorption peaks in an infra-red spectrum. Some candidates wrongly stated that the absorption peaks are the result of the vibration of a particular functional group instead of chemical bond. In part (c)(ii)(2), about half of the candidates were able to deduce one respective functional group that may be present in **Q** and **R**. In part (c)(iii), a small number of candidates were able to deduce and draw a possible structure of **Q**. Many candidates were unable to make use of the information provided to suggest that **R** is not a carboxylic acid and **Q** is a secondary alcohol.

School-based Assessment

All school candidates sitting for HKDSE Chemistry are required to participate in School-based Assessment (SBA). In 2025, the Chemistry SBA adopted the full resumption arrangement. A total of 13 227 students from 444 schools submitted their SBA marks. With the experience acquired over the past years, the implementation was generally smooth in most of the participating schools.

Based on the assessment data and samples of students' work submitted by participating schools, students' performance was in general satisfactory and within the expectations of the assessment requirements. To address potential discrepancies in the marking standard among individual teachers and schools, moderation on the SBA scores submitted by schools was performed according to both statistical method and professional judgment. It was reported that 51.4% of schools fell into the 'within the expected range' category, while 29.5% of schools had marks higher than expected, and 19.1% of schools had marks lower than expected this year. It was observed that the majority of schools with deviations only differed slightly from the expected range. This implies that most of the teachers have a good understanding of SBA implementation, and hence the marking standards are generally appropriate.

To ensure that teachers have a good understanding of the requirements and the principles of the assessment methods of the SBA, a Chemistry SBA conference and group meetings are held every year. The conference and group meetings can provide teachers with general comments and a summary of the SBA implementation, and the latest updates of the SBA requirements and administrative operations. Teachers can make use of the resources and support introduced at the conference to integrate practical work into lessons. Furthermore, the Education Bureau and the Hong Kong Examinations and Assessment Authority provide training courses and resources for teachers to enhance their knowledge and skill as well as to build up confidence in implementing SBA in schools.

To provide continuing support for teachers and to ensure fair implementation of the SBA, two supervisors and 24 district coordinators were appointed to address enquiries from teachers about SBA implementation and ensure that schools were administering the SBA in accordance with the stipulated guidelines. Phone calls, email correspondences, district group meetings and school visits were conducted to establish close connections between the district coordinators and the teachers. The said communication channels among the supervisors, district coordinators and teachers can enhance mutual understanding of the SBA implementation in schools. Based on the feedback and suggestions obtained from various sources, teachers and students have a better understanding of the essence and the requirements of the SBA. Nonetheless, some comments and recommendations are given below so that further improvement to the implementation of SBA can be made:

1. Variety of Experiments
   Even though there is no strict stipulated requirement regarding the types of experiments selected for SBA tasks besides including volumetric analysis and qualitative analysis in the task list, it is definitely beneficial to students' learning if they are exposed to a wider variety of experiment types. It is noted that a very small number of schools did not include 'titration' exercises in their volumetric analysis experiments, which may not be beneficial to the development of the holistic laboratory-based learning experience. Experiments from different topics like 'Chemistry reactions and energy', 'Rate of reaction', 'Chemical equilibrium', 'Organic Synthesis' and 'Analytical chemistry' can be used as SBA tasks. Conducting these types of experiments can enrich students' practical experience as well as strengthen students' practical skills and analytical thinking skills. For qualitative analysis, detection of ions (cations and / or anions) is required, but not the detection of organic species (e.g. distinguishing alkane and alkene). Moreover, 'Other experiments' refers to experiments suggested in the Curriculum and Assessment Guide, but not volumetric analysis or detection of ions. It is encouraging to observe that some schools, with sufficient lesson time and other resources, arranged 'preparative experiments' for their students.

2. Variety of Written Work
   Practical worksheets, experiment-related quizzes and simple / detailed laboratory reports, etc. are acceptable formats of written work in SBA. Teachers generally designed these tasks in a professional manner. Moreover, it is encouraging that most students can follow the instructions given by teachers in accomplishing the written work. Although there are no stipulated requirements in the SBA guidelines regarding the types of written work, writing detailed laboratory reports is very important for students studying science. Organising and writing a laboratory report in the correct format as well as presenting the data and experimental findings properly are very important skills.